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MATH 110

Week#3
2/6/26

Number: 3

Concept of Infinity

What is Infinity?

- I think Infinity means there is a limitless supply of something that continues to grow forever. For example, we could say time is infinite because theoretically, time will continue to pass, even after our absence. I like to think of infinity as a ladder that continues to go up, and disappears behind you as you climb up it.

Arithmetic with Remainders (Modular Arithmetic)

- How many times does 7 go into 1?

0 times with remainder 1

$1 = 0 \cdot 7 + 1$

$2 = 0 \cdot 7 + 2$

my # $3 = 0 \cdot 7 + 3$ remainder

$7 = 1 \cdot 7 + 0$

$8 = 1 \cdot 7 + 1$

$18 = 2 \cdot 7 + 4$

Remainder Mod 3

$18 = 6 \cdot 3 + 0$

$3 = 1 \cdot 3 + 0$ (Scribe)

Infinity Spokesperson

- limitless supply, continues to grow forever.
- immeasureable, large quantity, different types of ∞:
  - countable
  - uncountable
- symbol for something that lasts forever, can't be manipulated (ocean waves)

Ex. $l = 7$. 7 is the smallest natural # w/out a prime factorization.

- None of 2, 3, 4, 5, 6 divide 7. Thus the only divisors of 7 are itself & 1.
- Thus, 7 is prime, and $7 = 7$ is a prime factorization.

Ex. $l = 24$. 24 is the smallest natural # w/out a prime factorization.

- 2, 3, 4, 6, 8, 12. 24 is not prime, it has divisors other than 1 & 24.

$d = 4$ & $c = 6$.

$24 = 4 \cdot 6$ since 4 & 6 > 24, they have prime factorizations.

- $4 = 2 \cdot 2$, $6 = 2 \cdot 3$
- $24 = 4 \cdot 6 = 2 \cdot 2 \cdot 2 \cdot 3$. This is a prime factorization of 24.
- Thus, 24 isn't the smallest # w/ out a prime factorization.

Theorem: There are ∞ many prime #'s.

Proof: Suppose there are only finitely many prime #'s. I can list them all:

$$\overset{2}{p_1}, \overset{3}{p_2}, \overset{5}{p_3}, \overset{7}{\cdots} \overset{11}{p_n}.$$

- Let $Q = p_1 \cdot p_2 \cdot p_3 \cdot \cdots \cdot p_n = \prod_{i=1}^{n} p_i$

- Consider $Q+1$. We just proved $Q+1$ has a prime factorization.
- In particular, let z be a prime dividing $Q+1$. Since we listed all these primes, z is in the list $\{p_1, p_2, \cdots, p_n\}$
- Thus z divides Q
- But $z / Q+1$. Thus $z / (Q+1-Q)$.
  $z / 1$.
- This is a contradiction.
  The only divisor of 1 is 1, and 1 is NOT prime.
  Thus there are ∞ many primes. $\square$